

Problems

1. **(5 points)**. Three coins were tossed a total of 240 times, and each time the number of heads turning up was observed. The results are shown in Table 1, together with the results expected under the hypothesis that the coins are fair. Test this hypothesis at a significance level of 0.05.

Table 1: Result of Coin Tossing Experiment

	0 Heads	1 Head	2 Heads	3 Heads
Observed frequency	24	108	95	93
Expected frequency	30	90	90	30

import numpy as np

observed = np.array([24, 108, 95, 23])

expected = np.array([30, 90, 90, 30])

- Observed frequency: actual results from coin tossing experiment
- Expected frequency: values if the coin is fair

There are 4 possible heads outcomes (0,1,2,3) from the data, so: $4 - 1 = 3$

Significance level = 0.05

$$\chi^2_{0.05, 3} = 7.815$$

Final Answer: $\chi^2 = 6.711 < 7.815$

Therefore, we fail to reject the null hypothesis

At the 5% significance level, there is no statistical evidence to conclude that the coins are unfair.

Github link:

<https://github.com/jensathensg/jens/blob/f7f8aab45d9a31596e54a2ee08e8e74d6c54df17/1.py>

Results:

```
Chi-square value: 6.711111111111111
Fail to reject H0 (coins are fair)

[Execution complete with exit code 0]
```

2. **(5 points)**. Table 2 shows the numbers of students in each of two classes, *A* and *B*, who passed and failed an examination given to both groups. Using the (a) 0.05 and (b) 0.01 significance levels, test the hypothesis that there is no difference between the two classes. Work the problem with and without Yates' correction.

Table 2: Examination Results for *Classes A* and *B*

	Passed	Failed
<i>Class A</i>	72	17
<i>Class B</i>	64	23

Hypotheses:

- Null Hypothesis: There is no difference between Class A and Class B.
- Alternative Hypothesis: There is a difference between the two classes.

Critical Values:

At $\alpha = 0.05$: 3.841

At $\alpha = 0.01$: 6.635

Decision:

At 0.05

- Without Yates: Fail to reject H_0
- With Yates: Fail to reject H_0

At 0.01

- Without Yates: Fail to reject H_0
- With Yates: Fail to reject H_0

Githublink:

<https://github.com/jensathensg/jens/blob/a6cde79c0c634586749ec84800062848a94afa68/2.py>

Results:

```
Chi-square (Without Yates): 1.3480350373392294
Chi-square (With Yates): 0.9626911593773511
```

```
Decision at 0.05:
```

```
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Without Yates: Fail to reject H0
With Yates: Fail to reject H0
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```

```
Decision at 0.01:
```

```
-----
Without Yates: Fail to reject H0
With Yates: Fail to reject H0
-----
```

3. **(5 points).** Table 3 shows the relationship between hair and eye color of a sample of 200 students. (a) Find the coefficient of contingency without and with Yates' correction. (b) Compare the result of part (a) with the maximum coefficient of contingency.

Table 3: Relationship Between Hair Color and Eye Color

		Blonde	Not blonde
Eye Color	Blue	49	25
	Not Blue	30	96

Github link:

<https://github.com/jensathensg/jens/blob/f73d8c72349bd52a813cf92a509a17220dc2953d/3.py>

Results:

Final Answer: Since the computed values are less than the critical values, we fail to reject the null hypothesis.

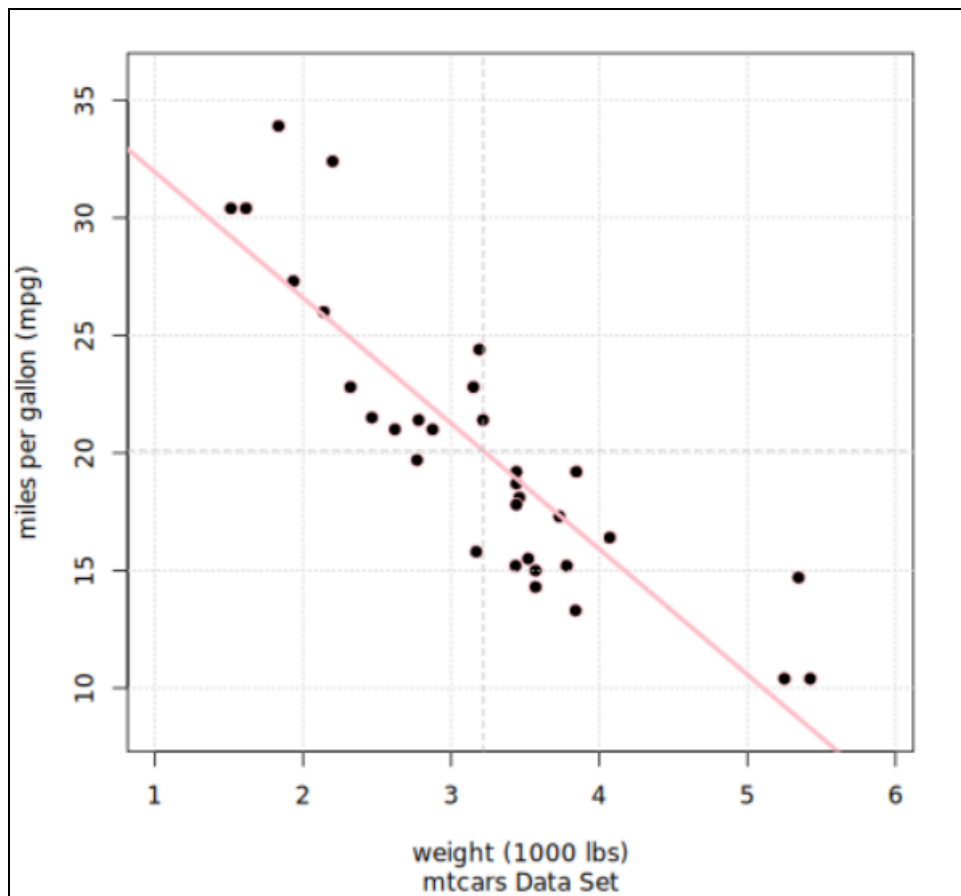
4. **(7 points).** The mtcars dataset in R contains measurements on 32 automobiles, including miles per gallon (mpg) and weight (wt, in 1000 lbs).
- Using the full dataset, fit a least-squares regression line with wt as the independent variable and mpg as the dependent variable.
 - Fit a second least-squares regression line with mpg as the independent variable and wt as the dependent variable.
 - Plot the data points and both regression lines on the same set of coordinate axes.
 - Interpret the slope of each regression line.

Instruction: Use R to load the dataset: `data(mtcars)`

Github link:

<https://github.com/jensathensg/jens/blob/6d3def6242249a169ed044cef48bf20fe35a6fd/4.r>

Results:



5. **(8 points)**. The cars dataset in R contains measurements of stopping distance (dist, in feet) and speed (speed, in miles per hour) for 50 automobiles.
- Using the full dataset, construct a scatter plot of stopping distance (D) against speed (V).
 - Fit a least-squares quadratic model of the form $D = a_0 + a_1V + a_2V^2$.
 - On the same graph from part (a), overlay the fitted quadratic regression curve.
 - Estimate the stopping distance when $V = 45$ mi/h and 80 mi/h.
 - Fit a simple linear regression model and overlay its fitted line on the same plot. Compare the two models based on their fit.

Instruction: Use R to load the dataset: `data(cars)`

Github link:

<https://github.com/jensathensg/jens/blob/3bf7de7819f6ebf8ab60dc47fb363e358bb87d4e/5.r>

Results:

